PROJEKT Maria Sibolt

Andmestiku analüüs: Data Science töökuulutused - Data Science Job Posting on Glassdoor

Eesmärk on uurida Data Science töökuulutuste andmestikku ja saada ülevaade ametitest, oskustest ja palkadest.

In [167…
```
import pandas as pd
import matplotlib.pyplot as plt
```

Andmestiku fail/faili lugemine

In [168…
```
file_path = 'Cleaned_DS_Jobs.csv'
data = pd.read_csv(file_path)
```

Andmestiku kirjeldus sisaldab:

- Ridade ja veergude arv
- Veergude nimed ja andmetüübid
- Puuduvate väärtuste arv
- Näidis esimestest ridadest

In [169…
```
print("ANDMESTIKU KIRJELDUS")
print(f"Ridade arv: {data.shape[0]}")
print(f"Veergude arv: {data.shape[1]}")
print("\nVeerud ja andmetüübid:")
print(data.dtypes)
print("\nPuuduvate väärtuste arv veergude lõikes:")
print(data.isnull().sum())
print("\nNäide esimestest ridadest:")
print(data.head())
```

ANDMESTIKU KIRJELDUS
Ridade arv: 660
Veergude arv: 27

Veerud ja andmetüübid:
Job Title   object
Salary Estimate   object
Job Description   object
Rating   float64
Company Name   object
Location   object
Headquarters   object
Size   object
Type of ownership   object
Industry   object
Sector   object
Revenue   object
min_salary   int64
max_salary   int64
avg_salary   int64
job_state   object
same_state   int64
company_age   int64
python   int64
excel   int64
hadoop   int64
spark   int64
aws   int64
tableau   int64
big_data   int64
job_simp   object
seniority   object
dtype: object

Puuduvate väärtuste arv veergude lõikes:
Job Title   0
Salary Estimate   0
Job Description   0
Rating   0
Company Name   0
Location   0
Headquarters   0
Size   0
Type of ownership   0
Industry   0
Sector   0
Revenue   0
min_salary   0
max_salary   0
avg_salary   0
job_state   0
same_state   0
company_age   0
python   0
excel   0
hadoop   0

```
spark                0
aws                  0
tableau              0
big_data             0
job_simp             0
seniority            0
dtype: int64

Näide esimestest ridadest:
           Job Title Salary Estimate  \
0  Sr Data Scientist        137-171
1     Data Scientist        137-171
2     Data Scientist        137-171
3     Data Scientist        137-171
4     Data Scientist        137-171

                                     Job Description  Rating  \
0  Description\n\nThe Senior Data Scientist is re...     3.1
1  Secure our Nation, Ignite your Future\n\nJoin ...     4.2
2  Overview\n\n\nAnalysis Group is one of the lar...     3.8
3  JOB DESCRIPTION:\n\nDo you have a passion for ...     3.5
4  Data Scientist\nAffinity Solutions / Marketing...     2.9

         Company Name       Location            Headquarters  \
0         Healthfirst   New York, NY            New York, NY
1             ManTech  Chantilly, VA             Herndon, VA
2      Analysis Group     Boston, MA              Boston, MA
3              INFICON     Newton, MA  Bad Ragaz, Switzerland
4  Affinity Solutions   New York, NY            New York, NY

                      Size        Type of ownership  \
0   1001 to 5000 employees   Nonprofit Organization
1  5001 to 10000 employees         Company - Public
2   1001 to 5000 employees  Private Practice / Firm
3    501 to 1000 employees         Company - Public
4      51 to 200 employees        Company - Private

                               Industry  ... company_age python  excel  \
0                   Insurance Carriers  ...          27      0      0
1                Research & Development  ...          52      0      0
2                            Consulting  ...          39      1      1
3  Electrical & Electronic Manufacturing  ...          20      1      1
4               Advertising & Marketing  ...          22      1      1

   hadoop  spark aws  tableau  big_data        job_simp  seniority
0       0      0   1        0         0  data scientist     senior
1       1      0   0        0         1  data scientist         na
2       0      0   1        0         0  data scientist         na
3       0      0   1        0         0  data scientist         na
4       0      0   0        0         0  data scientist         na

[5 rows x 27 columns]
```

In [170…

```
data = data.rename(columns={'avg_salary': 'keskmine_palk', 'job_simp': 'amet
```

Analüütilised päringud

1. Ametite järjestus kuulutuste järgi

In [171…
```
df = job_count = data['amet'].value_counts().reset_index()
job_count.columns = ['amet', 'count']
df.index = df.index + 1
print("\nPopulaarseimad ametid:")
df
```

```
Populaarseimad ametid:
```

Out[171…

	amet	count
1	data scientist	447
2	na	68
3	analyst	55
4	data engineer	46
5	mle	34
6	manager	7
7	director	3

2. Ametid kõrgeima keskmise palga järgi

In [172…
```
df = avg_salary_by_job = data.groupby('amet')['keskmine_palk'].mean().reset_
print("\nAmetid kõrgeima keskmise palgaga:")
df
```

```
Ametid kõrgeima keskmise palgaga:
```

Out[172…

	amet	keskmine_palk
4	manager	137.857143
6	na	127.088235
3	director	127.000000
2	data scientist	125.266219
5	mle	118.058824
0	analyst	115.527273
1	data engineer	113.782609

3. Oskuste nõudlus (nt Python, AWS, Spark)

In [173…
```
skills = ['python', 'aws', 'spark', 'excel', 'hadoop']
skill_counts = {skill: data[skill].sum() for skill in skills}
print("\nOskuste nõudlus:")
for skill, count in skill_counts.items():
    print(f"{skill.capitalize()}: {count}")
```

```
Oskuste nõudlus:
Python: 482
Aws: 172
Spark: 186
Excel: 291
Hadoop: 140
```

4. Keskmine palk ettevõtte suuruse järgi

In [174…
```
df = data.groupby('Size')['keskmine_palk'].mean().reset_index().sort_values(
print("\nKeskmine palk ettevõtte suuruse järgi:")
df
```

```
Keskmine palk ettevõtte suuruse järgi:
```

Out[174…

	Size	keskmine_palk
8	Unknown	143.176471
0	-1	130.666667
7	51 to 200 employees	127.648438
5	5001 to 10000 employees	126.442623
2	10000+ employees	122.632911
3	1001 to 5000 employees	121.538462
6	501 to 1000 employees	120.727273
1	1 to 50 employees	119.869048
4	201 to 500 employees	119.000000

5. Pythoni oskuse nõudluse protsent

In [175…
```
python_jobs = data['python'].value_counts(normalize=True) * 100
python_jobs_df = pd.DataFrame({'vajalik': python_jobs.index.map({1: 'Jah', 0
print("\nProtsent ametitest, kus on vaja Pythoni oskust:")
python_jobs_df
```

```
Protsent ametitest, kus on vaja Pythoni oskust:
```

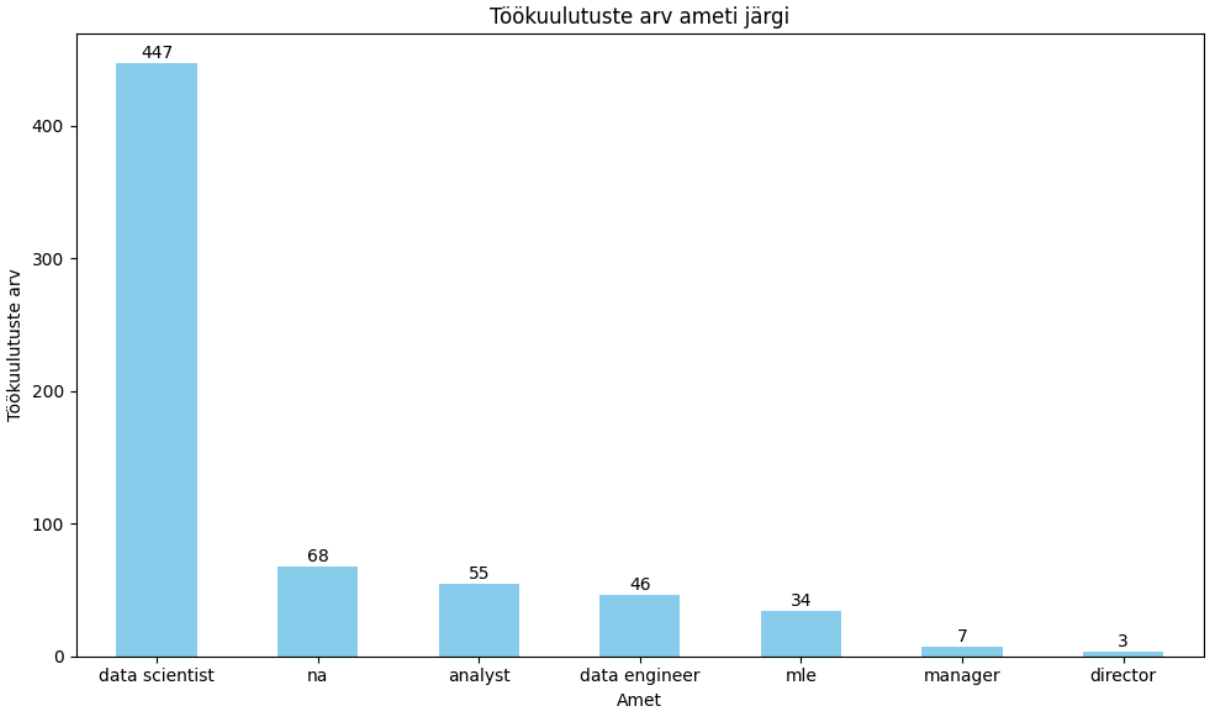
Out[175…

	vajalik	protsent
0	Jah	73.03
1	Ei	26.97

Diagrammid

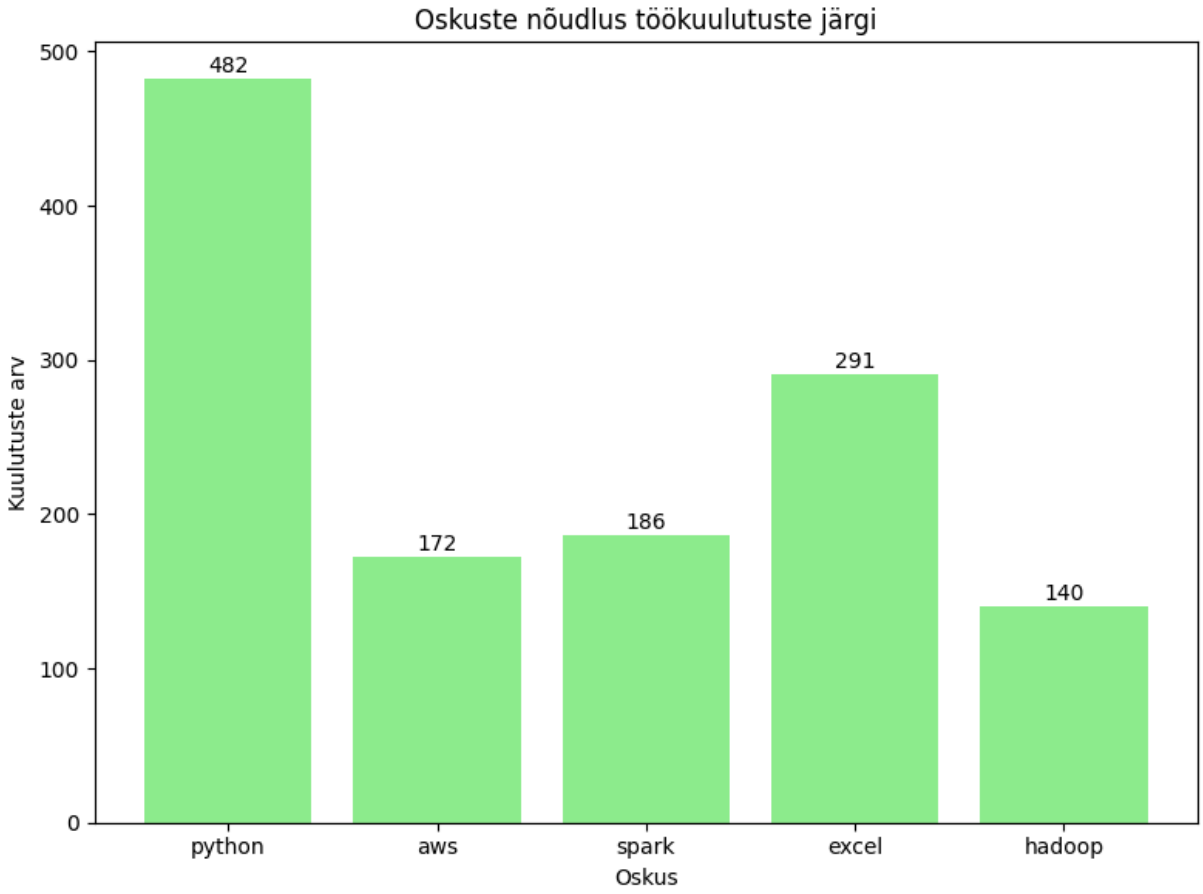
1. Töökuulutuste arv ameti järgi

In [176…
```
job_count = data['amet'].value_counts()
plt.figure(figsize=(10,6))
job_count.head(10).plot(kind='bar', color='skyblue')
plt.title('Töökuulutuste arv ameti järgi')
plt.ylabel('Töökuulutuste arv')
plt.xlabel('Amet')
plt.xticks(rotation=0)
for idx, value in enumerate(job_count.head(10)):
    plt.text(idx, value + 1, str(value), ha='center', va='bottom')
plt.tight_layout()
plt.show()
```



2. Oskuste nõudlus töökuulutuste järgi

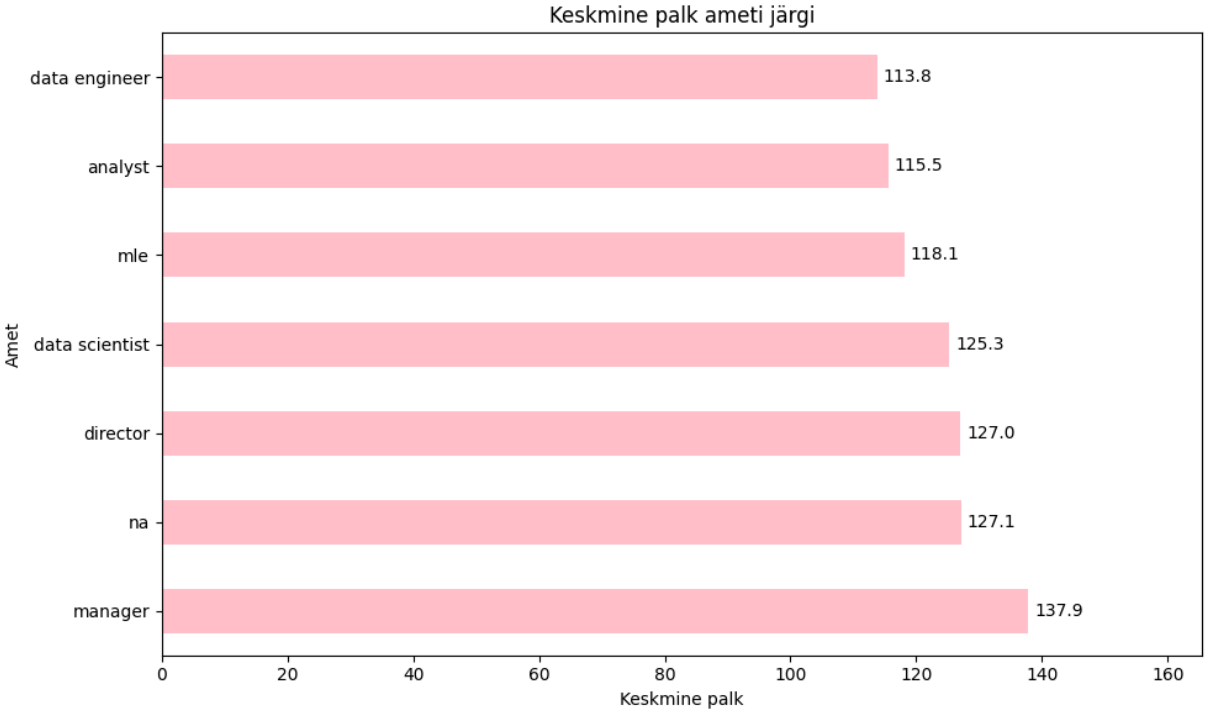
In [177…
```
plt.figure(figsize=(8,6))
bars = plt.bar(skill_counts.keys(), skill_counts.values(), color='lightgreer
plt.title('Oskuste nõudlus töökuulutuste järgi')
plt.ylabel('Kuulutuste arv')
plt.xlabel('Oskus')
for bar in bars:
    plt.text(bar.get_x() + bar.get_width() / 2, bar.get_height() + 1, str(ir
plt.tight_layout()
plt.show()
```



3. Keskmine palk ameti järgi

In [178…

```
avg_salary_by_job = data.groupby('amet')['keskmine_palk'].mean().sort_values
plt.figure(figsize=(10,6))
avg_salary_by_job.head(10).plot(kind='barh', color='pink')
plt.title('Keskmine palk ameti järgi')
plt.xlabel('Keskmine palk')
plt.ylabel('Amet')
for idx, value in enumerate(avg_salary_by_job):
    plt.text(value + 1, idx, f"{value:.1f}", va='center')
plt.xlim(0, avg_salary_by_job.max() * 1.2)
plt.tight_layout()
plt.show()
```



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